

Lab 4 plus

*Lab 4 plus is a combination of Lab 4 and an extra activity on ARP.

Packet Tracer Simulation – Exploration of ARP and Switch Table Communications

Objectives

- To explore ARP and switching operations.

Introduction

The topology is given to you. All IP addresses have been assigned to all devices. Please follow each step in sequence.

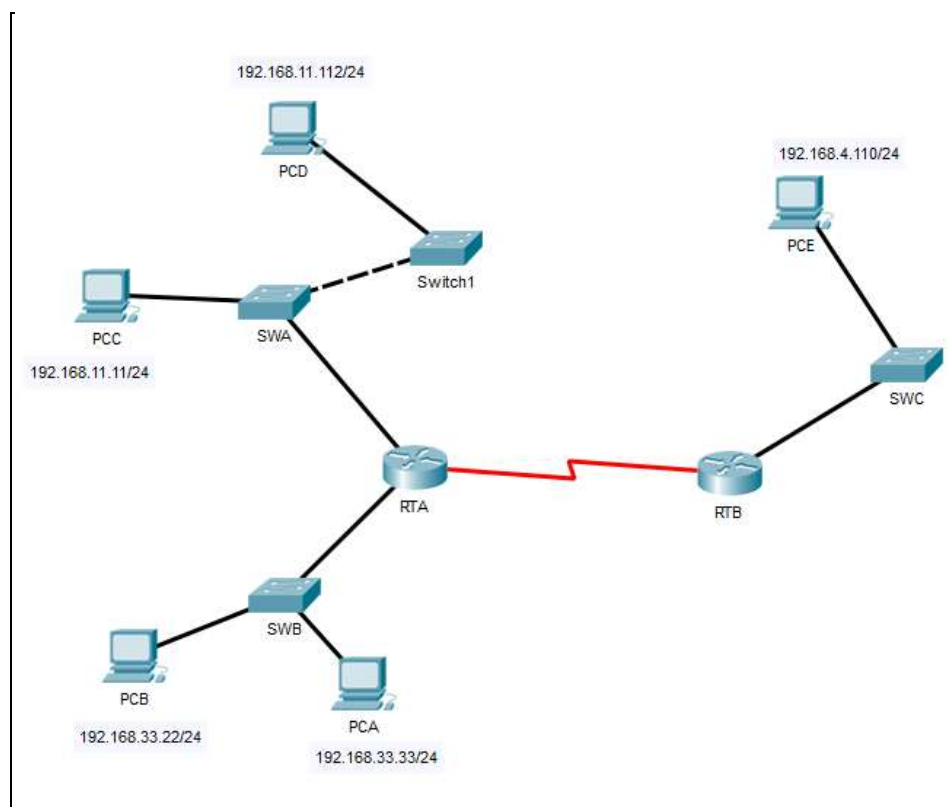


Figure 1

Part 1: Review the topology

Step 1: Open the *Lab_4_ARP.pkt*, and Perform the following tasks.

- a. At Router RTA, enter the CLI. At the command prompt type the following commands.
Snap the results after the last command and paste it here.

```
RTA>enable
RTA#show arp
```

```
RTA>enable
RTA#show arp
Protocol  Address           Age (min)  Hardware Addr  Type   Interface
Internet  192.168.11.1      -          0002.4A00.0E91  ARPA   FastEthernet1/0
Internet  192.168.33.1      -          000C.CF0C.593A  ARPA   FastEthernet0/0
```

- b. At Router RTB, enter the CLI. At the command prompt type the commands as in Figure 2. Snap the results after the last command and paste it here.

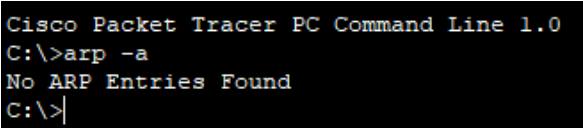
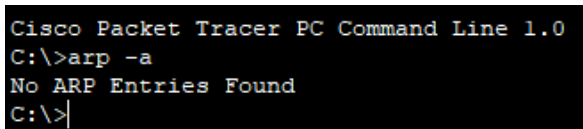
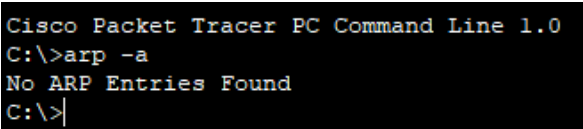
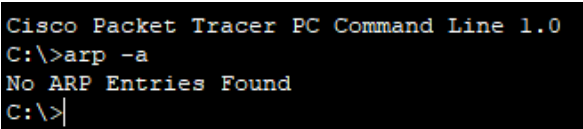
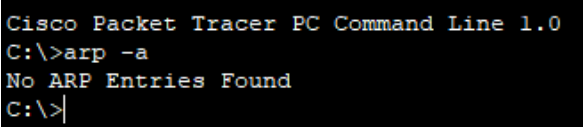
```
RTB>enable
RTB#show arp
Protocol  Address           Age (min)  Hardware Addr  Type   Interface
Internet  192.168.4.1       -          0001.977A.B614  ARPA   FastEthernet0/0
```

- c. At Switches SWA, SWAB and SWC, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
SWA>enable
SWA#show arp
SWA#show mac-address-table
```

SWA	<pre> SWA>enable SWA#show arp SWA#show mac-address-table Mac Address Table ----- Vlan Mac Address Type Ports ---- - 1 0002.4a00.0e91 DYNAMIC Fa0/1 1 000c.8546.7d85 DYNAMIC Fa1/1 SWA# </pre>
SWB	<pre> SWB>enable SWB#show arp SWB#show mac-address-table Mac Address Table ----- Vlan Mac Address Type Ports ---- - 1 000c.cf0c.593a DYNAMIC Fa0/1 SWB# </pre>
SWC	<pre> SWC>enable SWC#show arp SWC#show mac-address-table Mac Address Table ----- Vlan Mac Address Type Ports ---- - 1 0001.977a.b614 DYNAMIC Fa0/1 SWC# </pre>

- d. At PCA, click on the PC icon, and then choose Desktop-Command Prompt. At the command prompt type **arp -a** and click enter. Snap the results after the last command and paste it here. Do this to all PCs in the topology.

PCA 	PCB 
PCC 	PCD 
PCE 	

- e. What are your thoughts on the results?

No entry for all PCs because all the PCs didn't receive any ARP request. There will be no entry in the PC ARP table.

Part 2: Generate Network Traffic

Step 1: Generate traffic between PCA and PCB.

In the command prompt Perform the following tasks task to reduce the amount of network traffic viewed in the simulation.

- Click **PCA** and click the Desktop tab > Command Prompt.
- Enter the **ping 192.168.33.22** command. This may take a few seconds.
- In the Command prompt of PCA, type **arp -a**. Paste the result of this command here.
- In the Command prompt of PCB, type **arp -a**. Paste the result of this command here
- In the Command prompt of PCC, PCD abd PCE, type **arp -a**. Paste the result of this command here.

ping

```
C:\> ping 192.168.33.22

Pinging 192.168.33.22 with 32 bytes of data:

Reply from 192.168.33.22: bytes=32 time<1ms TTL=128
Reply from 192.168.33.22: bytes=32 time<1ms TTL=128
Reply from 192.168.33.22: bytes=32 time<1ms TTL=128
Reply from 192.168.33.22: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.33.22:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

PCA

```
C:\>arp -a

Internet Address      Physical Address      Type
192.168.33.22         0060.47ea.a746       dynamic
```

PCB

```
C:\>arp -a

Internet Address      Physical Address      Type
192.168.33.33         0002.1755.9a06       dynamic
```

PCC

```
C:\>arp -a

No ARP Entries Found
```

PCD

```
C:\>arp -a  
No ARP Entries Found
```

PCE

```
C:\>arp -a  
No ARP Entries Found
```

Step 2: Generate traffic between PCC to all other PC except PCA.

- Click **PCC** and click the Desktop tab > Command Prompt.
- Enter the **ping 192.168.33.22** command (ping to PCB). Then type **arp -a**. Paste the result after these commands here.

```
C:\>ping 192.168.33.22  
  
Pinging 192.168.33.22 with 32 bytes of data:  
  
Request timed out.  
Reply from 192.168.33.22: bytes=32 time<1ms TTL=127  
Reply from 192.168.33.22: bytes=32 time<1ms TTL=127  
Reply from 192.168.33.22: bytes=32 time<1ms TTL=127  
  
Ping statistics for 192.168.33.22:  
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),  
    Approximate round trip times in milli-seconds:  
        Minimum = 0ms, Maximum = 0ms, Average = 0ms  
  
C:\>arp -a  
Internet Address      Physical Address      Type  
192.168.11.1          0002.4a00.0e91       dynamic
```

- c. Enter the **ping 192.168.11.112** command (ping to PCD). Then type **arp -a**. Paste the result after these commands here.

```
C:\>ping 192.168.11.112

Pinging 192.168.11.112 with 32 bytes of data:

Reply from 192.168.11.112: bytes=32 time<1ms TTL=128
Reply from 192.168.11.112: bytes=32 time<1ms TTL=128
Reply from 192.168.11.112: bytes=32 time=4ms TTL=128
Reply from 192.168.11.112: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.11.112:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 4ms, Average = 1ms

C:\>arp -a

Internet Address      Physical Address      Type
192.168.11.1          0002.4a00.0e91        dynamic
192.168.11.112        0001.6462.0278        dynamic
```

- d. Enter the **ping 192.168.4.110** command (ping to PCE). Then type **arp -a**. Paste the result after these commands here.

```
C:\>ping 192.168.4.110

Pinging 192.168.4.110 with 32 bytes of data:

Request timed out.
Reply from 192.168.4.110: bytes=32 time=9ms TTL=126
Reply from 192.168.4.110: bytes=32 time=8ms TTL=126
Reply from 192.168.4.110: bytes=32 time=15ms TTL=126

Ping statistics for 192.168.4.110:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 8ms, Maximum = 15ms, Average = 10ms

C:\>arp -a

Internet Address      Physical Address      Type
192.168.11.1          0002.4a00.0e91        dynamic
192.168.11.112        0001.6462.0278        dynamic
```

- e. Discuss the results you got from all the commands on PCC.

The ARP entries are revealed in the ARP table only after PCC pings PCB, PCD, and PCE. The ARP table displays entries once PCB, PCD, and PCE receive the ARP request from PCC and subsequently receive pings from PCB, PCD, and PCE.

- f. At Router RTA, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
RTA>enable
RTA#show arp
```

```
RTA>enable
RTA#show arp
Protocol  Address           Age (min)  Hardware Addr  Type   Interface
-----
Internet  192.168.11.1      -          0002.4A00.0E91  ARPA   FastEthernet1/0
Internet  192.168.11.11     23         00D0.D39A.C0D9  ARPA   FastEthernet1/0
Internet  192.168.33.1      -          000C.CF0C.593A  ARPA   FastEthernet0/0
Internet  192.168.33.22     23         0060.47EA.A746  ARPA   FastEthernet0/0
RTA#
```

- g. At Router RTB, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
RTB>enable
RTB#show arp
```

```
RTB>enable
RTB#show arp
Protocol  Address           Age (min)  Hardware Addr  Type   Interface
-----
Internet  192.168.4.1       -          0001.977A.B614  ARPA   FastEthernet0/0
Internet  192.168.4.110     24         0060.702D.7C08  ARPA   FastEthernet0/0
```


Step 3: Switch MAC address table.

- a. At Switch SWA, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
SWA>enable
SWA#show arp

SWA#show mac-address-table
```

```
SWA>enable
SWA#show arp

SWA#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
1       0002.4a00.0e91   DYNAMIC Fa0/1
1       000c.8546.7d85   DYNAMIC Fa1/1
```

- b. At Switch SWB, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
SWB>enable
SWB#show arp

SWB#show mac-address-table
```

```
SWB>enable
SWB#show arp

SWB#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
1       000c.cf0c.593a   DYNAMIC Fa0/1
```

- c. At Switches SWC and Switch1, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
SWC>enable
SWC#show arp

SWC#show mac-address-table
```

```
SWC>enable
SWC#show arp

SWC#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
      1,    0001.977a.b614    DYNAMIC Fa0/1
```

- d. Do switches use arp table? (Y/N)

Y

- e. Explain your answer in (d) **Hint: the answer may surprise you. Google for the explanation. It is not part of NetComm syllabi, it is just for knowledge..*

Layer 3 switches use ARP tables because they operate at both Layer 2 and Layer 3, enabling them to resolve IP addresses to MAC addresses for routing within a subnet. However, traditional Layer 2 switches do not use ARP tables, as they rely solely on MAC address tables for forwarding frames at the Data Link Layer.

- f. What information does the command `show mac-address-table` gives?

VLAN: The VLAN associated with the learned MAC address.

MAC Address: The specific MAC address that the switch has learned.

Type: Whether the MAC address was learned dynamically or manually configured (static).

Ports: The port on the switch where the MAC address was learned or is associated.

Part 3: Attach wireless lab results.

In this part, you will use **Lab_4_Wireless.pka** file.

Step1: Change the filename of the pka file.

- Change the Lab 4 filename to include your name. *Example: *Lab4AliAhmad.pkt*
- Go through the instructions. As you complete the tasks, you will see the bottom right hand corner of the pkt file increase in completion percentage, until you get 100/100.

The screenshot displays the Packet Tracer interface. On the left, a network topology is shown with a central switch (S1) connected to three PCs (PC1, PC2, PC3) and a wireless router (WR52). The switch has three subinterfaces: G0/0.10 (VLAN 10), G0/0.20 (VLAN 20), and G0/0.88 (VLAN 88). PC1 is connected to S1 G0/0.10, PC2 to S1 G0/0.20, and PC3 to WR52. The wireless router is connected to S1 G0/0.88. The activity wizard window on the right is titled "Packet Tracer - Configuring Wireless LAN Access". It contains an "Addressing Table" with the following data:

Device	Interface	IP Address	Subnet Mask	Default Gate
R1	G0/0.10	172.17.10.1	255.255.255.0	N/A
	G0/0.20	172.17.20.1	255.255.255.0	N/A
	G0/0.88	172.17.88.1	255.255.255.0	N/A
PC1	NIC	172.17.10.21	255.255.255.0	172.17.10.1

At the bottom of the activity wizard, the "Completion" status is shown as "100/100" and "1/1".

- Once you have completed fully, capture the screen (which includes the filename, the topology and the activity wizard showing completion) and paste it here.

Cisco Packet Tracer - D:\Users\Nbtu\Documents\STUDY\YEAR 2\Network Communications\Lab 4\Lab_4_LFE_VIN_SHEN.pkt - Guest - 2025-01-24 15:56:01

File Edit Options View Tools Extensions Window Help

Logical Physical 1:277, 8:7

Subinterfaces

- G0/0/10 172.17.10.1/24
- G0/0/20 172.17.20.1/24
- G0/0/88 172.17.88.1/24

PC1
172.17.10.2/24
VLAN 10

PC2
172.17.20.2/24
VLAN 20

PC3
172.17.88.2/24
VLAN 88

WRS2
172.17.88.25/24
VLAN 88

S1
F0/18

PT Activity: 00:16:47

preconfigured to establish a connection with the router.

- Click PC3 > Desktop > PC Wireless.
- Click the Profiles tab.
- Click **New**.
- Name the new profile **Wireless Access**.
- On the next screen, click **Advanced Setup**. Then manually enter the SSID of **WRS_LAN** on **Wireless Network Name**. Click **Next**.
- Choose **Obtain network settings automatically (DHCP)** as the network settings, and then click **Next**.
- On **Wireless Security**, choose **WPA2 Personal** as the method of encryption and click **Next**.
- Enter the passphrase **cisco123** and click **Next**.
- Click **Save** and then click **Connect to Network**.

Step 2: Verify PC3 wireless connectivity and IP addressing configuration.

- The **Signal Strength** and **Link Quality** indicators should show that you have a strong signal.
- Click **More Information** to see details of the connection including IP addressing information.
- Close the PC Wireless configuration window.

Part 3: Verify Connectivity

All the PCs should have connectivity with one another.

Time Elapsed: 00:16:47 Completion: 100/100

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